Vyze

Offline AI Vision Assistant

See for those who cannot. Everywhere. Instantly. Privately.

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THE PROBLEM

01 A World Designed for the Sighted

Over 2.2 billion people worldwide live with vision impairment (WHO, 2026). Of these, at least 1 billion could have been prevented or remain untreated. By 2050, this number will increase by 55% -- adding 600 million more people.

> *"Every 5 seconds, someone in the world goes blind."*
>
> - International Agency for the Prevention of Blindness

For the 61,000+ registered visually impaired individuals in Malaysia alone, daily tasks that sighted people take for granted -- reading a label, crossing a street, identifying currency, recognizing a friend -- remain constant challenges.

02 Why Existing Solutions Fall Short

Limitation	Impact on Daily Life
Requires Internet	Fails underground, in rural areas, on planes, in developing regions
Cloud Latency	1-3 second delay per query makes real-time navigation impossible
Privacy Risk	Camera images of home, medication, documents uploaded to third-party servers
Subscription Cost	Monthly fees create financial barriers for disabled users
No Memory	App never learns user's environment, routes, or preferences
English Only	Excludes 80% of the global visually impaired population

The result: visually impaired users are forced to choose between connectivity-dependent tools that fail when they need them most, and basic assistive devices that offer no intelligence at all.

THE SOLUTION

03 Vyze: AI That Sees for You -- Offline

Vyze is a fully offline AI vision assistant that speaks what the camera sees -- instantly, privately, and without ever connecting to the internet.

100% Offline

No internet needed. Works underground, on planes, in rural areas. Forever free.

Instant Response

First audio in under 1 second. Sentence-streamed TTS for real-time interaction.

Learns Over Time

Adaptive memory remembers your home, office, routes, and preferences.

Speaks Your Language

Auto-detects spoken language and responds in it. No manual switching.

04 How It Works

1	**SPEAK**	User asks a question naturally: "What medicine is this?" or "What's in front of me?"
2	**CAPTURE**	Camera captures a fresh frame from the live preview -- zero disk I/O, all in memory.
3	**READ**	ML Kit OCR extracts text in 80-150ms. Gemma 3n E2B interprets with full context.
4	**RESPOND**	Sentence-buffered TTS speaks the answer aloud as it generates -- no waiting.
5	**REMEMBER**	Interaction stored in local database. Next time, Vyze provides smarter context.

05 User Experience: A Day with Vyze

Morning -- Medicine

"What is this?" Vyze reads the prescription label: "This is Diclac Retard, a pain relief medication containing diclofenac sodium. Take one tablet daily after meals." Response time: 400ms.

Commute -- Navigation

Continuous mode auto-describes surroundings: "Bus stop ahead on your left, about 10 steps. Two people waiting. Sign reads RapidKL Route 300."

Office -- Document

"Read this." Vyze extracts text from a meeting agenda via ML Kit OCR in 150ms, then reads it aloud with contextual formatting.

Evening -- Shopping

"How much is this?" Vyze identifies the Malaysian ringgit banknote: "This is a 20 ringgit note." Instant, private, no internet.

TECHNOLOGY

06 What Powers Vyze

Gemma 3n E2B

Google's 2-billion parameter multimodal vision-language model, int4 quantized to 3.66 GB. Runs entirely on-device via LiteRT-LM with NPU/GPU acceleration.

ML Kit OCR

Google's on-device text recognition supporting Latin (English, Malay, Indonesian, Vietnamese) and Chinese scripts. 80-150ms extraction time.

Adaptive Memory

Room database with vector similarity search. Learns user's environment, routes, and preferences over time.

Language Mirroring

Auto-detects spoken language via Android SpeechRecognizer, mirrors to both AI output and TTS voice. No hardcoded language lists.

Sentence Streaming

Tokens stream directly to TTS as they generate. First audio fires after just 3 characters -- no waiting for full response.

07 Performance Benchmarks

Metric	Result	How
Text Query (OCR)	300-500ms	ML Kit + Gemma interpretation
Scene Description	1.5-2s	Gemma 3n E2B direct inference
First Audio	<1s	Sentence-buffered TTS streaming
Prefill Tokens	~80	Aggressively trimmed prompts
Image Size	256x256	Center-cropped, spatially aligned
Voice Detection	300ms	Aggressive silence endpoints
Model Loading	2-4s	GPU warm-up on first launch

MARKET OPPORTUNITY

08 The Market is Massive and Underserved

2.2 Billion

People worldwide with vision impairment (WHO, 2026)

$410.7B

Annual global productivity loss from sight loss (IAPB, 2024)

61,000+

Registered visually impaired in Malaysia (DOSM, 2024)

09 Target Segments

Blind & Low-Vision Users

Primary users. Need real-time scene understanding, text reading, and navigation assistance. 2.2 billion globally, 61,000+ in Malaysia.

Elderly Population

Age-related vision decline affects 27.8% of Malaysians over 50. Simple voice interface requires zero technical literacy.

Developing Regions

Areas with limited internet connectivity where cloud-based assistive tools simply don't work.

Multilingual Communities

Southeast Asia's diverse linguistic landscape needs tools that adapt to the user's language, not the other way around.

COMPETITIVE EDGE

10 Vyze vs. Existing Solutions

Feature	Vyze	Be My AI	Seeing AI	Envision	Lazarillo
Fully Offline	Yes	No	Partial	No	Partial
VLM on Device	Yes	No	No	No	No
Adaptive Memory	Yes	No	No	No	No
Multi-Language	Yes	No	Yes	Yes	Yes
Language Mirroring	Yes	No	No	No	No
OCR Speed	300ms	2-3s	1s	1.5s	N/A
First Audio	<1s	3-5s	2-3s	2-3s	1-2s
Cost	Free	Free*	Free*	Sub	Free
Privacy	Full	Cloud	Cloud	Cloud	Cloud

** Requires internet connection for core functionality*

Vyze is the only solution that combines fully offline VLM inference, adaptive memory, language mirroring, and sentence-streamed TTS -- all without any cloud dependency or subscription cost.

11 What Makes Vyze Defensible

Offline-First Architecture

While competitors race to add cloud features, Vyze's constraint is its strength. It works where nothing else does -- underground, in rural areas, on planes, in developing regions.

Adaptive Intelligence Loop

Every interaction makes Vyze smarter. The local memory system creates personalized spatial awareness that cloud competitors cannot replicate without violating privacy.

Hybrid OCR + VLM Pipeline

ML Kit for speed, Gemma for intelligence. This dual-engine approach delivers 10-30x faster text extraction than competitors.

Zero Recurring Costs

No API fees, no subscriptions, no data plans. The model runs on hardware the user already owns. This is the most sustainable business model for accessibility.

ROADMAP

12 Product Roadmap

Phase 1: Core Experience

- Currency detection (Malaysian ringgit banknotes)
- Offline OpenStreetMap navigation with GPS
- Multi-language OCR expansion (Hindi, Arabic, Thai)
- Medication reminder system with prescription OCR
- Scene history and visual diary

Phase 2: Intelligence Layer

- Smart glasses integration (Ray-Ban Meta, Envision Glasses)
- Real-time object tracking across frames
- Public transport announcements (offline GTFS feeds)
- Contact/face recognition for personal identification
- Indoor navigation with user-recorded floor plans

Phase 3: Scale & Impact

- Smaller model routing (fast model for simple queries)
- Federated learning for privacy-preserving pattern sharing
- Haptic navigation via smartwatch integration
- Emergency detection and auto-alert system
- Social media photo description sharing

13 Business Model

Vyze's core product will always be free and open-source. Sustainability comes from complementary revenue streams that don't compromise the accessibility mission.

Hardware Partnerships

Co-develop smart glasses and wearable accessories. Revenue from hardware sales, not software subscriptions.

Enterprise Licensing

Hospitals, rehabilitation centers, and care facilities deploy Vyze for patient assistance. Per-device enterprise licenses.

Government & NGO Grants

Disability rights organizations, WHO, and national health departments fund distribution to underserved communities.

Premium Features

Advanced features like multi-device sync, custom model training, and priority support for power users.

IMPACT

14 Why This Matters

> *"The real problem is not whether machines think, but whether men do."*
>
> - B.F. Skinner

Vyze is not just an app. It is a statement that artificial intelligence should serve everyone -- not just those with internet access, not just English speakers, not just those who can afford monthly subscriptions.

When a blind person in rural Malaysia can point their phone at a medicine label and instantly hear what it says -- in their language, privately, for free -- that is technology fulfilling its highest purpose.

Vyze makes the invisible visible. Offline. Instantly. For everyone.

Vyze

See for those who cannot.

Everywhere. Instantly. Privately.

github.com/moreese84/Vyze

Built for accessibility. Designed for independence.